CSS Flexbox

Flexbox (Flexible Box Layout) is a **one-dimensional layout system** used to arrange elements in a row or column efficiently.

It solves common layout problems like:

- Horizontal & vertical centering
- Equal height columns
- Space distribution between items
- Responsive alignment without float hacks

What is Flexbox?

Flexbox works with:

- **Flex Container** → Parent element (display: flex)
- **Flex Items** → Direct children of the flex container

```html
<div class="container">
    <div class="item">1</div>
    <div class="item">2</div>
    <div class="item">3</div>
</div>
```

```css
.container {
    display: flex;
}
```

Once you apply `display: flex`, the layout behavior changes completely.

Flex Container Properties

1. display: flex

Activates flexbox.

```
.container {
    display: flex;
}
```

2. flex-direction

Controls direction of items.

Values:

- `row` (default)
- `row-reverse`
- `column`
- `column-reverse`

🔹 Demo

```
.container {
    display: flex;
    flex-direction: column;
}
```

Items stack vertically.

3. justify-content (Main Axis Alignment)

Aligns items horizontally (default row).

Values:

- `flex-start`
- `flex-end`
- `center`
- `space-between`
- `space-around`
- `space-evenly`

🔹 Demo – Horizontal Center

```
.container {
    display: flex;
    justify-content: center;
}
```

4.align-items (Cross Axis Alignment)

Aligns items vertically (default row).

Values:

- `stretch`
- `flex-start`
- `flex-end`
- `center`
- `baseline`

🔹 Demo – Vertical Center

```
.container {
    display: flex;
    height: 200px;
    align-items: center;
}
```

5. flex-wrap

Controls wrapping.

Values:

- `nowrap` (default)
- `wrap`
- `wrap-reverse`

🔹 Demo

```
.container {
    display: flex;
    flex-wrap: wrap;
}
```

Flex Item Properties

1. flex-grow

Defines how much item grows.

```
.item1 {
    flex-grow: 1;
}
```

```
.item2 {
    flex-grow: 2;
}
```

Item2 takes double space.

2. flex-shrink

Defines shrink behavior.

✅ 3. flex-basis

Initial size before growing/shrinking.

```
.item {
    flex-basis: 200px;
}
```

✅ 4. flex (Shortcut)

```
.item {
    flex: 1; /* grow shrink basis */
}
```

Equivalent to:

```
flex: 1 1 0;
```